

Unit 3: Regression Analysis & Regularization

Handwritten Student Notebook

Syllabus Topics Covered:

CIE-1 AUTHORITATIVE SYLLABUS TOPICS COVERED Introduction to Regression: Dependent vs independent variables, Regression vs Classification, Performance evaluation metrics (MSE, RMSE, MAE, R^2 , Adjusted R^2). Simple Linear Regression: Model assumptions, Ordinary Least Squares (O...

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Introduction to Regression Analysis

§ 1

Regression analysis models the functional relationship between one or more **independent explanatory variables** (features $\mathbf{X} = [x_1, \dots, x_d]$) and a **continuous numerical dependent variable** (target Y).

Regression vs. Classification

- **Target Domain:** Classification outputs discrete categorical classes ($Y \in \{0, 1\}$ or $\{C_1, \dots, C_k\}$); Regression outputs real-valued continuous numbers ($Y \in \mathbb{R}$).
- **Evaluation Metrics:** Classification uses Accuracy, Precision, Recall, F_1 , and ROC-AUC; Regression uses Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2).

Quick Recall

Regression predicts continuous numerical quantities ($Y \in \mathbb{R}$). Evaluated via Mean Squared Error (MSE) and R^2 .

Linear Regression: Ordinary Least Squares (OLS)

§ 2

Linear regression assumes a linear relationship between features and target.

Simple Linear Regression (Single Feature)

$$y = mx + c + \epsilon$$

Ordinary Least Squares (OLS) minimizes the Residual Sum of Squares (RSS):

$$RSS = \sum_{i=1}^n (y_i - (mx_i + c))^2$$

Taking partial derivatives with respect to m and c and setting them to zero yields the closed-form solutions:

OLS Closed-Form Formulas:

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$c = \bar{y} - m\bar{x}$$

Multiple Linear Regression (Matrix Normal Equation)

For d features, $\hat{y} = \mathbf{X}\mathbf{w}$. The closed-form analytical solution that minimizes Mean Squared Error is the **Normal Equation**:

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})(y_i - \bar{y})$	$(y_i - \bar{y})^2$
1	2	-2	-2	4	+4	4
2	3	-1	-1	1	+1	1
3	5	0	+1	0	0	1
4	4	+1	0	1	0	0
5	6	+2	+2	4	+4	4
Total	0	0	0	10	9	10

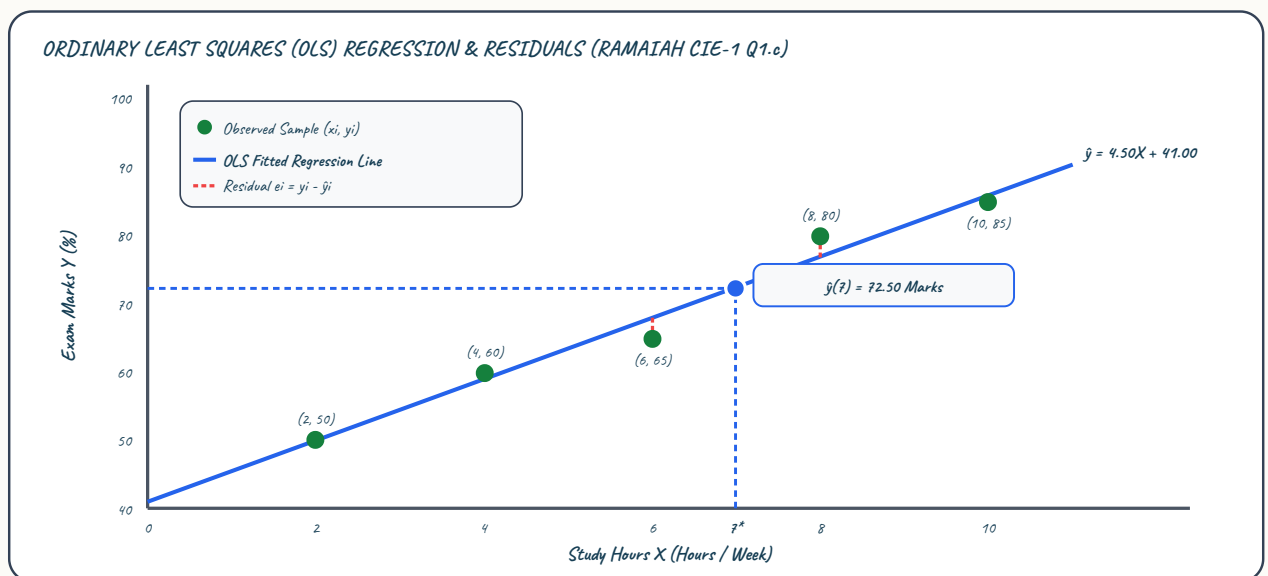


Figure 3.1: Ordinary Least Squares (OLS) Regression Fit & Residuals (Ramaiah CIE-1 Q1.c Solved)

Study Hours (X) vs Exam Marks (Y) yields optimal least-squares line $\hat{Y} = 4.50X + 41.00$. Vertical red lines depict residuals $e_i = y_i - \hat{y}_i$. Target prediction for $X=7$ hours is $\hat{Y}(7) = 72.50$ marks.

[Interactive Visualizer Online]: Interactive Linear Regression & Error Minimizer. Explore the step-by-step visualizer on the live course notes page.

Quick Recall

OLS slope $m = \frac{\text{Cov}(X,Y)}{\text{Var}(X)}$ and $c = \bar{y} - m\bar{x}$. Matrix Normal Equation: $\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$. $R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$.

Multiple Linear Regression & Normal Equations

§ 3

When a continuous outcome y depends on multiple predictors $x_1, x_2, \dots, x_p$, **Multiple Linear Regression** generalizes simple OLS into matrix algebra.

Matrix Formulation of Multiple Regression:

For n observations and p features, the model is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\mathbf{y} \in \mathbb{R}^{n \times 1}$, design matrix $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ (with leading column of 1s for intercept), coefficient vector $\boldsymbol{\beta} \in \mathbb{R}^{(p+1) \times 1}$, and residuals $\boldsymbol{\epsilon} \in \mathbb{R}^{n \times 1}$.

The Normal Equations (Closed-Form Analytical Solution)

Minimizing the sum of squared errors $S(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ by setting the gradient $\frac{\partial S}{\partial \boldsymbol{\beta}} = \mathbf{0}$ yields the classic **Normal Equations**:

$$\mathbf{X}^T \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^T \mathbf{y} \implies \boldsymbol{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Assumption & Invertibility:

For $(\mathbf{X}^T \mathbf{X})^{-1}$ to exist, matrix $\mathbf{X}$ must have full column rank ($p + 1$). If features are linearly dependent (exact multicollinearity), $(\mathbf{X}^T \mathbf{X})$ becomes singular and cannot be inverted without regularization (Ridge regression).

Worked Numerical: Two-Predictor House Price Prediction (SEE 2026 Q5.a)

Predict house price (Y , in Lakhs) from Size (X_1 , hundreds of sq.ft) and Bedrooms (X_2):

House	Size (X_1)	Bedrooms (X_2)	Price (Y , Lakhs)
H1	12	2	50
H2	15	3	65
H3	18	3	80
H4	20	4	90

Solving via least squares yields estimated model: $\hat{Y} = -6.67 + 4.17X_1 + 3.33X_2$.

Holding bedrooms fixed, each additional 100 sq.ft of area adds Rs. 4.17 Lakhs to the predicted

Polynomial Regression & Bias-Variance Tradeoff

§ 4

When the relationship between features and target is non-linear (e.g., parabolic or cyclical), fitting a straight line causes severe **underfitting (high bias)**. **Polynomial Regression** extends linear regression by introducing powers of features as new independent variables:

$$\hat{y} = w_0 + w_1x + w_2x^2 + w_3x^3 + \dots + w_dx^d$$

Crucially, polynomial regression remains a **linear model** because it is linear in its parameters $\mathbf{w}$, and can be solved directly using the standard OLS Normal Equation over the transformed feature matrix $\Phi(X)$.

The Bias-Variance Dilemma Across Polynomial Degrees

Degree (d)	Model Behavior	Bias vs. Variance	Remedy
Degree 1 (Linear)	Underfitting: straight line fails to capture curvature	High Bias, Low Variance	Increase polynomial degree
Degree 2–3 (Optimal)	Optimal generalization: captures underlying trend without fitting noise	Balanced Bias and Variance	Cross-validation model selection
Degree 15+ (Excessive)	Overfitting: oscillates wildly to pass through every individual noise point	Low Bias, High Variance	Regularization (L_2 Ridge / L_1 Lasso)

Quick Recall

Polynomial regression models non-linear phenomena via $\sum w_k x^k$. It is linear in parameters. High degree polynomials cause overfitting (high variance), mitigated via Ridge/Lasso regularization.

Regularization in Regression: Ridge (L_2) & Lasso (L_1)

§ 5

When features are collinear or numerous (*ppproxn*), Ordinary Least Squares suffers from high variance (overfitting). **Regularization** constrains parameter magnitudes by adding a penalty term to the loss function.

Property	Ordinary Least Squares (OLS)	Ridge Regression (L_2)	Lasso Regression (L_1)
Penalty Term	None	$\lambda \sum_{j=1}^p \beta_j^2 = \lambda \ \boldsymbol{\beta}\ _2^2$	$\lambda \sum_{j=1}^p \beta_j = \lambda \ \boldsymbol{\beta}\ _1$
Objective Function	$\min \frac{1}{2n} \ \mathbf{y} - \mathbf{X}\boldsymbol{\beta}\ ^2$	$\min \left[\text{MSE} + \lambda \ \boldsymbol{\beta}\ _2^2 \right]$	$\min \left[\text{MSE} + \lambda \ \boldsymbol{\beta}\ _1 \right]$
Analytical Solution	$\boldsymbol{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$	$\boldsymbol{\beta} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$	No closed form; solved via Coordinate Descent
Sparsity / Feature Selection	No	No (shrinks coefficients toward zero, never exactly 0)	Yes (sets coefficients to exactly zero, eliminating features)
Computational Benefit	Can be singular if multicollinear	Guarantees $(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})$ is invertible	Produces compact, interpretable models

Faculty Problem Note (Shruti Mam's Problem Set):

In Ridge regression, increasing λ increases bias but dramatically reduces model variance. For $\lambda = 0$, Ridge reverts to standard OLS. As $\lambda \rightarrow \infty$, all $\beta_j \rightarrow 0$, predicting the horizontal mean $\bar{y}$.

Gradient Descent Optimization for Regression

§ 6

When the number of features p is large (e.g. $p > 10,000$), computing the analytical matrix inverse $(\mathbf{X}^T \mathbf{X})^{-1}$ with time complexity $O(p^3)$ is computationally prohibitive. **Gradient Descent** provides an iterative optimization alternative.

Batch, Stochastic, and Mini-Batch Gradient Descent

- **Batch Gradient Descent:** Computes the gradient of MSE over the entire dataset n at each step. Convergence is smooth and monotonic, but slow for large datasets.
- **Stochastic Gradient Descent (SGD):** Computes the gradient using a single random training sample at each step. Fast updates with noisy trajectory; can escape shallow local minima.

- **Mini-Batch Gradient Descent:** Computes the gradient over a small subset (e.g. $B = 32, 64, 128$). Combines the stability of batch descent with hardware vectorization speedups (GPU/SIMD).

Update Rule: $\text{oldsymbol}^{(t+1)} = \text{oldsymbol}^{(t)} - \text{alpha} \nabla_{\text{oldsymbol}} J(\text{oldsymbol})$, where alpha is the learning rate hyperparameter.

Logistic Regression

§ 7

Despite its name, **Logistic Regression** is a **classification algorithm** used to estimate the probability that an instance belongs to a binary class ($Y \in \{0, 1\}$).

The Sigmoid (Logistic) Function:

Linear regression maps input features to unbounded real values: $z = \mathbf{w}^T \mathbf{x} \in (-\infty, +\infty)$. Logistic regression passes z through the **Sigmoid Activation Function** $\sigma(z)$ to squash outputs into valid probabilities in $(0, 1)$:

$$P(Y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x})}}$$

Properties: $\sigma(0) = 0.5$, $\sigma(z) \rightarrow 1$ as $z \rightarrow +\infty$, and $\sigma(z) \rightarrow 0$ as $z \rightarrow -\infty$.

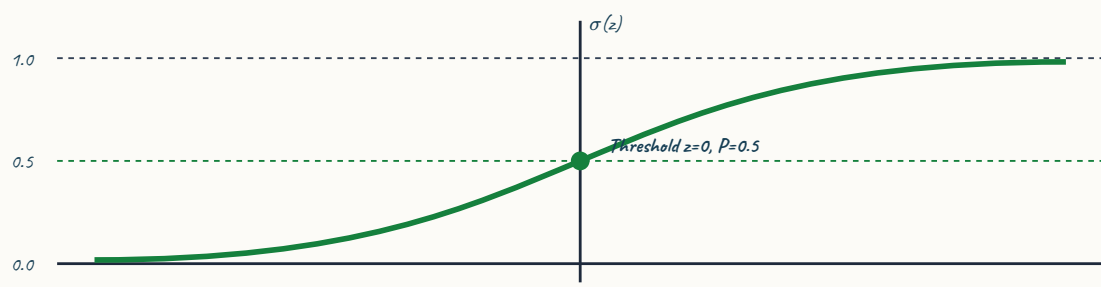


Figure 3.2: The logistic sigmoid curve mapping the real line into calibrated probabilities.

Odds Ratio and the Logit Function

The **Odds** of an event is the ratio of probability of occurrence to non-occurrence:

$$\text{Odds} = \frac{p}{1-p} = e^z$$

Taking the natural logarithm yields the **Logit**:

$$\ln \left(\frac{p}{1-p} \right) = z = w_0 + w_1 x_1 + \dots + w_d x_d$$

Thus, logistic regression is linear in the log-odds space!

Loss Function: Binary Cross-Entropy (Log Loss)

We cannot use Mean Squared Error for logistic regression because squaring $\sigma(z)$ creates a non-convex loss surface plagued by local minima. Instead, we use Maximum Likelihood Estimation, leading to the convex **Binary Cross-Entropy Loss**:

$$J(\mathbf{w}) = -\frac{1}{n} \sum_{i=1}^n \left[y_i \ln(\hat{p}_i) + (1 - y_i) \ln(1 - \hat{p}_i) \right]$$

Solved Problem 4 · Exam Level Logistic Classification

A trained binary logistic regression model for predicting cardiovascular risk has the following logit equation:

$$z = -3.0 + 0.5x_1 + 0.2x_2$$

Where x_1 is daily cigarette consumption and x_2 is patient body mass index (BMI).

1. Compute the logit value z for a patient with $x_1 = 4$ cigarettes and $x_2 = 10$ BMI units above baseline.
2. Compute the odds of developing cardiovascular disease for this patient.
3. Compute the predicted probability $P(Y = 1 \mid \mathbf{x})$.
4. Using a classification threshold of 0.5, predict the diagnostic class.

Step 1: Compute Logit z

$$z = -3.0 + 0.5(4) + 0.2(10) = -3.0 + 2.0 + 2.0 = +1.0000$$

Step 2: Compute Odds

$$\text{Odds} = e^z = e^{1.0} \approx 2.7183$$

The patient is approximately 2.72 times more likely to be at risk than healthy.

Step 3: Compute Predicted Probability $\sigma(z)$

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-1.0}} = \frac{1}{1 + 0.367879} = \frac{1}{1.367879}$$

Step 4: Classification Decision

Since $P(Y = 1 \mid \mathbf{x}) = 0.7311 \geq 0.50$, the patient is classified into **Class 1 (High Risk)**.

Final Verified Results Logit $z = +1.0000$ | Odds = 2.7183 | $P(Y=1) = 0.7311$ (73.11%) →
Classified as Positive (1)

Quick Recall

Logistic regression models probability via $\sigma(z) = \frac{1}{1+e^{-z}}$. Linear in log-odds $\ln \frac{p}{1-p} = \mathbf{w}^T \mathbf{x}$. Optimized via Binary Cross-Entropy (Log Loss).

Unit 3 Quick Revision Sheet (Regression Only)

High-yield formula cheat sheet and algorithm summaries for rapid CIE-1 exam review.

Simple OLS Regression

$$b_1 = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2}$$

$$b_0 = \bar{Y} - b_1 \bar{X}$$

$$\hat{Y} = b_0 + b_1 X$$

$$R^2 = 1 - \frac{SSE}{SST}$$

Multiple Linear Regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

$$\boldsymbol{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$R^2_{\text{adj}} = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

Ridge Regularization (L2)

$$\text{Loss: } \text{extMSE} + \lambda \sum \beta_j^2$$

$$\boldsymbol{\beta}_{\text{extRidge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

Shrinks weights; never zero.

Lasso Regularization (L1)

$$\text{Loss: } \text{extMSE} + \lambda \sum |\beta_j|$$

Enforces sparsity: sets weights to 0.

Performs feature selection.

Gradient Descent Updates

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \alpha \nabla J(\boldsymbol{\beta}^{(t)})$$

Batch: full n ; SGD: 1 sample;

Mini-batch: size B (e.g. 32, 64).

Logistic Classification

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\text{extOdds} = \frac{p}{1 - p} = e^z$$

$$\text{extLogit}(p) = \ln(\text{extOdds}) = \mathbf{w}^T \mathbf{x} + b$$

Loss: Binary Cross-Entropy

Important Examination Questions (Regression)

Curated questions structured according to VTU and Ramaiah examination standards.

2-Mark / 3-Mark QuestionsPractice Problem:

Q1. Differentiate between Linear Regression and Logistic Regression.

Solution:

Linear Regression: Models continuous numerical targets $y \in \mathbb{R}$ using unbounded linear hypotheses $\hat{y} = \mathbf{w}^T \mathbf{x} + b$, evaluated via MSE.

Logistic Regression: Models categorical class probabilities $P(y = 1 | \mathbf{x}) \in (0, 1)$ using the non-linear Sigmoid activation function, evaluated via Binary Cross-Entropy (Log-Loss).

Practice Problem:

Q2. Define the Sigmoid function and state its range and mathematical derivative.

Solution:

The Sigmoid activation function is defined as $\sigma(z) = \frac{1}{1 + e^{-z}}$. It squashes any real-valued input from $(-\infty, +\infty)$ into the probability range $(0, 1)$. Its derivative is computationally efficient: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.

Practice Problem:

Q3. Why is Adjusted R^2 preferred over standard R^2 in Multiple Regression?

Solution:

Standard R^2 mechanically increases or remains constant whenever a new predictor is added to a model, even if the feature is useless random noise. Adjusted R^2 introduces a degrees-of-freedom penalty for every additional parameter added: $R^2_{\text{adj}} = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$, ensuring it increases only if the new feature genuinely improves predictive accuracy.

5-Mark / 6-Mark Questions

Practice Problem:

Q4. Compare Ridge Regression (L_2) and Lasso Regression (L_1) with their loss functions and properties.

Solution:

Ridge Regression: Adds an L_2 squared penalty $\lambda \sum \beta_j^2$. Has closed-form solution $\beta = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$. Shrinks coefficients smoothly toward zero but never forces them to exact zero.

Lasso Regression: Adds an L_1 absolute value penalty $\lambda \sum |\beta_j|$. Due to the sharp corners of the L_1 constraint diamond, it forces irrelevant coefficients to become identically zero, performing automatic embedded feature selection.

Practice Problem:

Q5. Explain the concept of Odds Ratio and Log-Odds (Logit function) in Logistic Regression.

Solution:

Odds: The ratio of probability of occurrence to probability of non-occurrence:

$$\text{Odds} = \frac{p}{1-p} = e^z.$$

Logit: Taking the natural logarithm of odds yields the logit function: $\text{Logit}(p) =$

$$\ln\left(\frac{p}{1-p}\right) = z = \mathbf{w}^T \mathbf{x} + b.$$

This shows that Logistic Regression is a generalized linear model that assumes the log-odds of the positive class are a linear combination of the input predictors.

8-Mark / 10-Mark Questions

Practice Problem:

Q6. Derive the Ordinary Least Squares (OLS) closed-form formulas for slope b_1 and intercept b_0 in simple linear regression.

Solution:

Objective: Minimize the Sum of Squared Residuals: $S(\eta_0, \eta_1) = \sum_{i=1}^n (y_i - (\eta_0 + \eta_1 x_i))^2$.

1. Partial derivative with respect to η_0 :

$$\frac{\partial S}{\partial \eta_0} = -2 \sum (y_i - \eta_0 - \eta_1 x_i) = 0 \implies \sum y_i - n\eta_0 - \eta_1 \sum x_i = 0$$

$$\text{Dividing by } n: \bar{y} - \eta_0 - \eta_1 \bar{x} = 0 \implies \eta_0 = \bar{y} - \eta_1 \bar{x}.$$

2. Partial derivative with respect to η_1 :

$$\frac{\partial S}{\partial \eta_1} = -2 \sum x_i (y_i - \eta_0 - \eta_1 x_i) = 0 \implies \sum x_i y_i - \eta_0 \sum x_i - \eta_1 \sum x_i^2 = 0$$

Substitute $\eta_0 = \bar{y} - \eta_1 \bar{x}$:

$$\sum x_i y_i - (\bar{y} - \eta_1 \bar{x}) \sum x_i - \eta_1 \sum x_i^2 = 0 \implies \sum x_i y_i - \bar{y} \sum x_i = \eta_1 (\sum x_i^2 - \bar{x} \sum x_i)$$

Using algebraic identities $\sum x_i y_i - n \bar{x} \bar{y} = \sum (x_i - \bar{x})(y_i - \bar{y})$ and $\sum x_i^2 - n \bar{x}^2 = \sum (x_i - \bar{x})^2$:

$$\eta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{1}{n} \sum x_i y_i - \bar{x} \bar{y}$$

Solved Questions from Past Department Examinations

ARCHIVE

Transcribed directly from official Ramaiah Institute of Technology papers in notes/ml/practice/ml-cie-1-2-see.pdf (CIE-1 March 2026, Course Code 23IS62) and faculty coursework:

Ramaiah Internal Assessment - I (March 2026) · Q 1(c) [3 Marks · Blooms L3, CO3]

"The following data represents hours studied (X) and marks obtained (Y): X = [2, 4, 6, 8, 10], Y = [50, 60, 65, 80, 85]. i. Compute slope (b) and intercept (a). ii. Write the regression equation. iii. Predict marks for X = 7 hours."

Step 1: Compute Means ($n = 5$):

$$\bar{X} = \frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6.0, \quad \bar{Y} = \frac{50 + 60 + 65 + 80 + 85}{5} = \frac{340}{5} = 68.0$$

Step 2: Compute Summations:

$$\sum X = 30, \quad \sum Y = 340, \quad \sum X^2 = 220, \quad \sum XY = (2)(50) + (4)(60) + (6)(65) + (8)(80) + (10)(85) = 2220$$

Step 3: Compute Slope (b) and Intercept (a):

$$b = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2} = \frac{5(2220) - (30)(340)}{5(220) - (30)^2} = \frac{11100 - 10200}{1100 - 900} = \frac{900}{200} = 4.5$$

$$a = \bar{Y} - b\bar{X} = 68.0 - (4.5)(6.0) = 68.0 - 27.0 = 41.0$$

Step 4: Regression Equation & Prediction:

$$\text{Regression Equation: } \hat{Y} = 41.0 + 4.5X$$

$$\text{For } X = 7 \text{ hours: } \hat{Y}(7) = 41.0 + 4.5(7) = 41.0 + 31.5 = 72.5$$

Ramaiah Internal Assessment - I (March 2026) · Q 2(c) [3 Marks · Blooms L2, CO3]

"Differentiate Logistic Regression and Linear Regression in machine learning."

- Output Target:** Linear Regression predicts continuous numeric quantities $y \in (-\infty, +\infty)$, whereas Logistic Regression predicts discrete probabilities $P(y = 1|\mathbf{x}) \in (0, 1)$ mapped to class labels $\{0, 1\}$.
- Hypothesis Function:** Linear regression uses $\hat{y} = \mathbf{w}^T \mathbf{x} + b$; logistic regression wraps the linear score inside the Sigmoid function $\sigma(z) = 1/(1 + e^{-z})$.
- Loss Function:** Linear regression minimizes Mean Squared Error (MSE), whereas logistic regression minimizes Binary Cross-Entropy (Log-Loss) because MSE is non-convex when combined with the sigmoid activation function.

Practice Problems (Step-by-Step Verified Solutions)

PRACTICE

The following problems are drawn from the faculty problem sheets (Linear_Regression and multiple linear regression_problems.docx and Linear & ridge regression....docx) with verified step-by-step arithmetic:

Problem 1: Simple Linear Regression · Ice Cream Sales vs Temperature Easy · Verified Formula

A shop records daily temperature ($^{\circ}\text{C}$) and corresponding ice cream sales (units) for 8 days:

$$X = [20, 22, 24, 26, 28, 30, 32, 34], Y =$$

$[120, 135, 150, 165, 180, 195, 210, 225]$. 1. Find the regression equation. 2. Predict sales when temperature is 29°C .

Summations ($n = 8$): $\sum X = 216 \implies \bar{X} = 27.00, \sum Y = 1380 \implies \bar{Y} = 172.50$
 $\sum X^2 = 6000, \sum XY = 38520$.

Slope: $b = \frac{8(38520) - (216)(1380)}{8(6000) - (216)^2} =$

$\frac{308160 - 298080}{48000 - 46656} = \frac{10080}{1344} = \mathbf{7.5000}$.

Intercept: $a = 172.50 - (7.5000)(27.00) = 172.50 - 202.50 = \mathbf{-30.0000}$.

Equation: $\hat{Y} = -30.0000 + 7.5000X$. For $X = 29^{\circ}\text{C}$: $\hat{Y}(29) = -30.0 + 7.5(29) = \mathbf{187.50\text{ units}}$.

Problem 2: Regularization · Ridge vs OLS Coefficient Shrinkage Moderate · Verified Matrix Math

Given centered 1D feature data where $\sum x_i^2 = 100$ and $\sum x_i y_i = 400$. Calculate the OLS slope and the Ridge slope for regularization penalty $\lambda = 20$.

OLS Slope: $\hat{\beta}_{\text{OLS}} = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{400}{100} = \mathbf{4.0000}$.

Ridge Slope ($\lambda = 20$): $\hat{\beta}_{\text{Ridge}} = \frac{\sum x_i y_i}{\sum x_i^2 + \lambda} = \frac{400}{100 + 20} = \frac{400}{120} = \mathbf{3.3333}$.

Conclusion: The Ridge penalty successfully shrunk the coefficient from 4.00 to 3.33 (a 16.7% shrinkage), dampening sensitivity to potential noise in the feature.